Sr.No. 8517

Bharatiya Vidya Bhavan's

SARDAR PATEL INSTITUTE OF TECHONOLOGY

(An Autonomous Institute Affiliated to University of Mumbai)
MUNSHI NAGAR, ANDHERI (WEST), MUMBAI - 400058

MID SEMESTER EXAMINATION

AY: 2025-26	Session: Odd MSE October 2025	Date: 08-10-2025
Sem: 1	Slot: 10:00am - 11:00am	C. Code: CS413
Exam Type: Written	Block:401 Course Section:1	Seat No: 2022600045

1_29_502_986_1_1505_38416

Name of the candidate. Vinay . Racha

Candidate's UID number: 2 0 2 2 6 0 0 0 4 5

Examination class room number: 4 0 1

Invigilator's signature with date: 8/10/25

(To be filled in by the candidate)

EXAMINATION: BE TECH MSE
(For example FY MCA/FY BTECH)

SEMESTER : VII

I / II / III / IV / V / VI / VII / VIII

PROGRAM: CSE-AIML

DAY AND DATE: WeD 8/10/25

COURSE NAME: Deep learning

(To be filled in by the examiner)

Question numbers	1	2	3	4	5	6	Total	Signature of the examiner
Marks awarded								
Marks after revaluation								

INSTRUCTIONS TO CANDIDATES

1) Candidate should ensure that all answer books including supplementary answers books received by them bear the signature of the junior supervisor, otherwise the answer books will not be examined.

2) Begin the answer to each question on a new page. For each answer, write the corresponding question number and sub question number if any in the respective column provided in the answer sheet.

3) Do not write anything in the column provided for the marks to be assigned by the examiners.

4) Candidate will not be permitted to leave the examination hall until an hour after the question papers are distributed.

5) Every candidate present must sign against their seat number on the attendance sheet provided by the junior supervisor.

6) Candidates are forbidden to (i) bring any book, notes. Scribbling papers, mobile telephones, programmable calculators, or any other similar devices.(ii) speak or communicate in any manner to any other candidate, while the examination is in progress, and (iii) take with them any answer-book written or blank while leaving the examination hall. The supervisors/authorized persons are authorized to check the students.

7) Candidate suspected to be guilty of any of the aforesaid acts will be allowed to write their paper only after giving an undertaking in writing that the decision of the College Examination Authorities in respect of the reported act of unfair means is binding on them.

8) Candidates should write their answers legibly. They are warned that **zero marks** will be assigned to **answers which cannot be assessed by the examiners owing to illegible handwriting.**

9) Write on both side of a page. Rough work, when necessary, should be done on the Left- hand side and in pencil only.

10) While underlining of answers for focusing attention is permitted, use of varied inks, except for illustration and figures must be avoided.

11) No sheet shall be torn from the answer-books provided nor shall additional papers attached to them.

12) The answer-books will be scrutinized before they are sent to examiners.

13) All answer-books supplied shall be returned whether written or blank.

14) Nothing shall be written on the question-paper.

15) If candidate needs anything, they should approach the Junior Supervisor without disturbing other candidates. However, they should not leave their seat on my account.

16) A warning bell will be given ten minutes before the close of the examination. Candidates will not be allowed to leave the examination hall during the last ten minutes. At the final bell, they must stop writing and be ready to hand over their answer-books to the Junior Supervisor. They should not leave their seats until answers-books from all candidates are collected by the Junior Supervisor.

17) A candidate who disobeys any instructions issued by the Senior/Junior Supervisor or who is guilty of rude or disobedient behavior is liable for disciplinary action to be taken against him/her by the College Examination Authorities.

Radhe Krishna

Sr. No. 8517

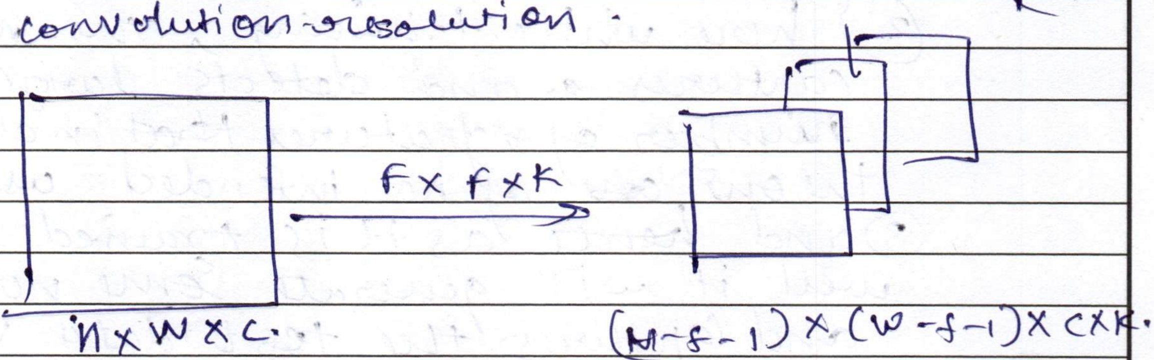
START WRITING HERE
(Begin answer for each question on a new page)

Q1a

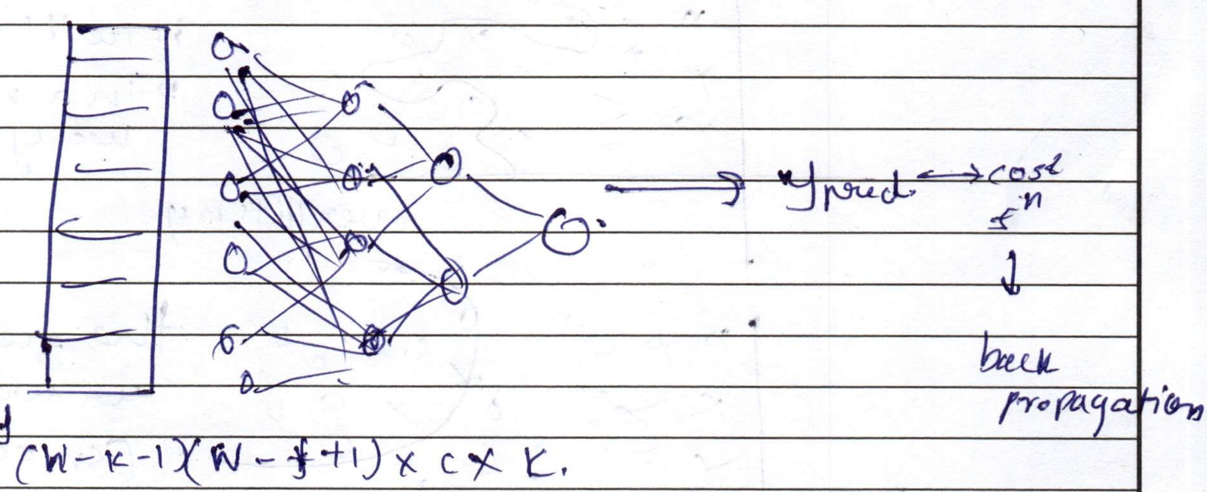
2a) 2D convolutions.

Input = H x W x C.
Filter k filters of size F x F.

convolution resolution:



~~These~~ thus will be filltered & passed to the fully connected network.



Duplicate separate convolutions are critical in morder architechture, because they have lesser parameters applying that can turned in less

Sr. No. 8517

Space For Marks	Question No.	START WRITING HERE (Begin answer for each question on a new page)

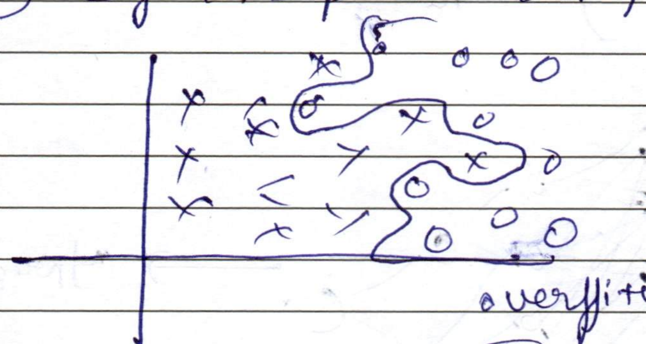
Q1 b) (1) overfitting: ~~occurs when~~ is a major problem in ML models

(2) It occurs when the model works very well on (training) Input data and perform poorly on the test data

(2) here while training, the model captures ~~a~~ and detects large number of ~~x~~ features ~~that~~ including the one ~~ass~~ of no intended use

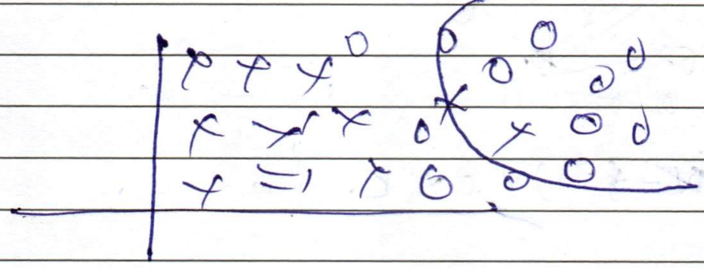
(3) and ~~hence~~ as it is trained so well it will generate some noise. and captures the test data ~~we~~ with very high accuracy hence this is the problem of overfitting

(4) eg to separate categories



It will separate it in a nonlinear way.

overfitting



the ~~aresult~~ result we wanted (smooth curve)

(5) when model is complex (overfitting) high variance occurs

(6) when model is simple (underfitted) high bias occurs.

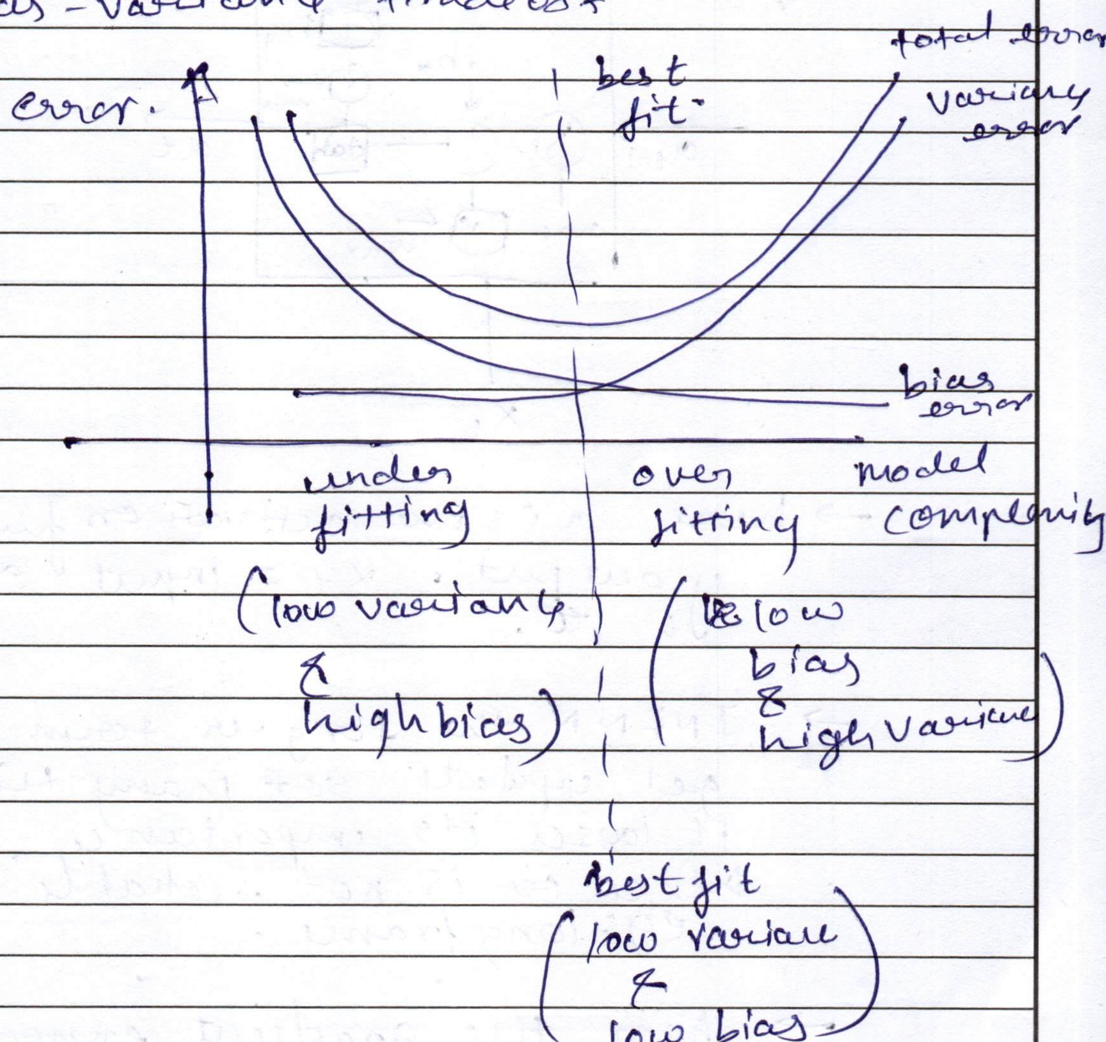
(7) we want a base error in the model i.e low variance & bias error. (best fit)

Sr. No. 8517

Space For Marks	Question No.	START WRITING HERE (Begin answer for each question on a new page)

(8) total error = (Bias Error)^2 + variance error + base error

(9) bias - variance tradeoff



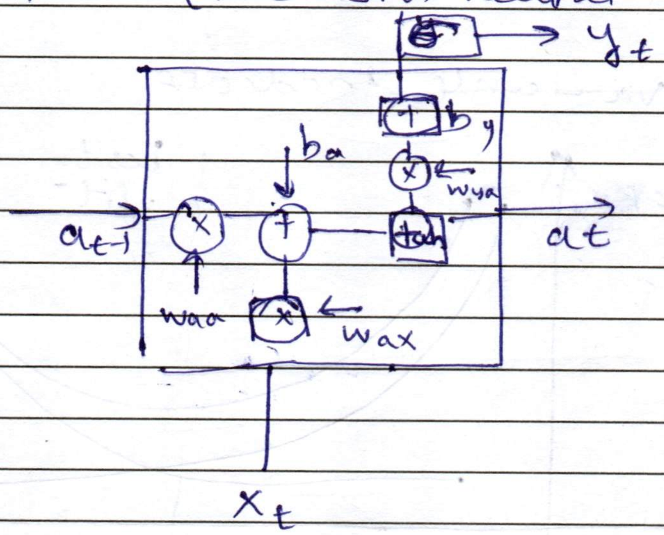
(10) conclusion hence when model is more complex it captures many features hence of overfitting occurs. i.e (model ~~cor~~ capacity is ~~too~~ high) and it ~~over~~ less generalise over captures the features. i.e less generalization.

Sr. No. 8517

SARDAR PATEL INSTITUTE OF TECHNOLOGY

Space For Marks	Question No.	START WRITING HERE (Begin answer for each question on a new page)

Q3a] → RNN (recursive neural network)



→ here a is the activation function
y_t output. x_t = input at time stamp ~~t~~. (t)

→ IN RNN the longer term memory get update ~~so~~ many times that it looses its importance
hence ~~cu~~ is not reliable for ~~lon~~ eg: long frame.

→ here the gradient expression of RNN = 0
(which is know as vanishing gradient problem) as the long term state is being updated so many time it losses its importance

→ & here $a_t = \tanh(w_{aa} a_{t-1} + w_{ax} x_t + b_a)$

$y_t = \text{softmax}(w_{ya} a_t + b_y)$

→ from Backpropagation, we know that:

$$\boxed{\frac{\delta L}{\delta h_{t-k}} = \sum_{t=1}^{t} \frac{\partial L}{\partial O} \times \frac{\partial O_t}{\partial a_t} \times \frac{\partial a_t}{\partial h_{t-k}}}$$

→

START WRITING HERE
(Begin answer for each question on a new page)

Sr. No. 8517

$o \rightarrow$ output
$a \rightarrow$ activation function

$$\therefore \frac{\partial L}{\partial h_{t-k}} = \sum_{t=1}^{t=T} \frac{\partial L}{\partial o} \times \frac{\partial o_t}{\partial a_t} \times \frac{\partial a_t}{\partial h_{t-k}}.$$

here if we solve this,

$$= \cancel{\frac{\partial L}{\partial o}}$$

$$= \frac{\partial L}{\partial o}\frac{\partial o_t}{\partial a_t} \times \frac{\partial a_t}{\partial h_{t-k}} + \frac{\partial L}{\partial o}\frac{\partial o_t}{\partial a_t}\frac{\partial o_{t-1}}{\partial a_{t-1}} \times \frac{\partial a_{t-1}}{\partial h_{t-k}}$$

$$+ \cdots \frac{\partial L}{\partial o}\underbrace{\left(\frac{\partial o_t}{\partial a_t} * \frac{\partial o_{t-1}}{\partial a_{t-1}} * \cdots \frac{\partial o_1}{\partial a_1}\right)}\frac{\partial o_1}{\partial h_{t-k}}.$$

as these are gradient value from (0 to 1) hence multiplying them, ≈ 0

$$\frac{\partial L}{\partial h_{t-k}} \approx 0.$$

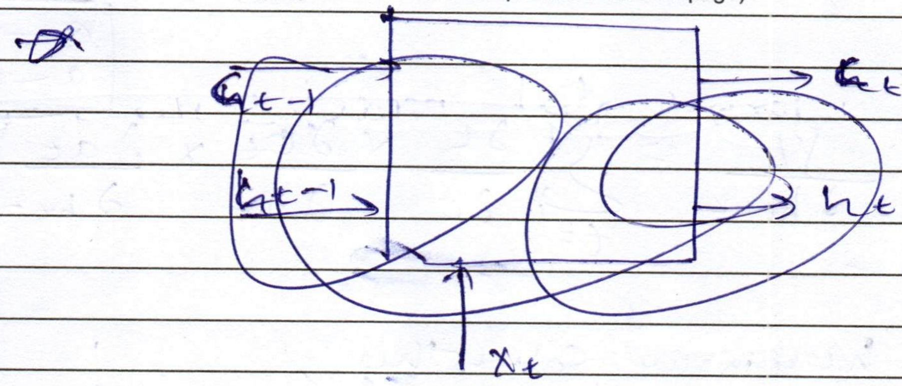
hence the gradient expression of RNN ≈ 0
i.e the vanishing gradient problem exist

→ LSTM
stands for long & short term memory. which solves the problem of vanishing gradient.
→ here we have 3 gates.
1. Forget gate f_t
2. input gate i_t
3. output gate o_t

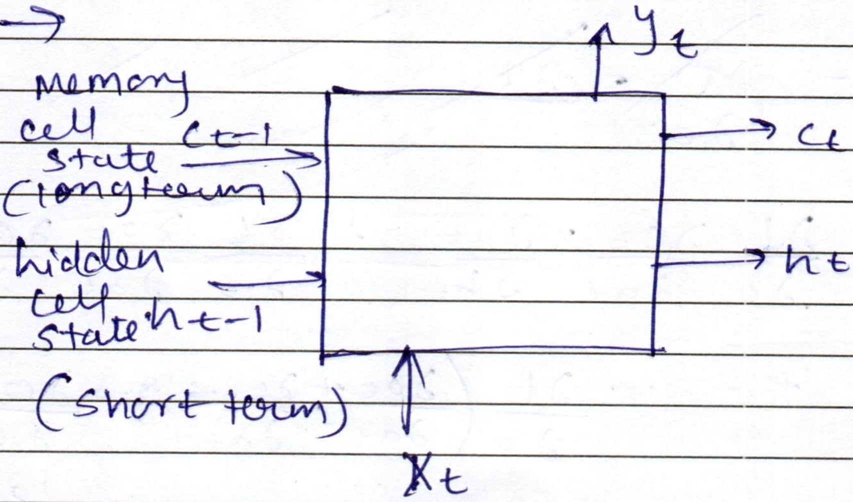
→

START WRITING HERE
(Begin answer for each question on a new page)

Space For Marks	Question No.	



→



→ here the forget gate is used to retain the ~~co~~ long term memory and also to ~~wa~~ what to forget

~~→ Iput & output gate, to~~

→ $\boxed{C_t = f_t \times C_{t-1} + i \times \tilde{C}}$

eg: $\begin{bmatrix} 0 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} - & - & - \\ - & - & - \\ - & - & - \end{bmatrix}$

here the forget gate will retain the information for $\underline{1}$ & for $\underline{0}$ it will forget

→ $\tilde{c}$ = candidate value

$\boxed{\tilde{c} = \tanh(W_c(h_{t-1} + X_t) + b_c)}$

i.e new ~~ing~~ input.

~~value~~ .

Sr. No. 8517

Space For Marks	Question No.	START WRITING HERE (Begin answer for each question on a new page)

and hence @ ~~memory~~ memory cell state,
the forget gate modifies the recursion
and prevent vanishing ~~gradient~~ gradient
~~issue~~ ..

Space For Marks	Question No.	START WRITING HERE (Begin answer for each question on a new page)

~~Q3b] → GRU (Gatted Recursion~~

Q3b] → GRU (Gatted recursion unit) is more or less similar in concept to LSTM. (but with only 1 state)

→ here the aim is also ~~to~~ to retain the longterm knowledge.

→ but here we have only 2 gates

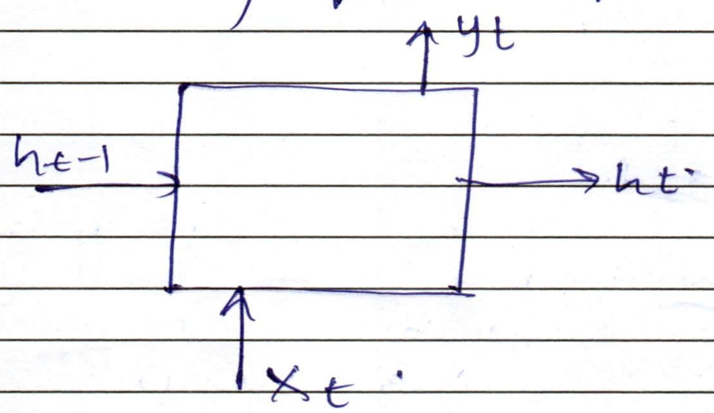
(1) update gate (U_t)

(2) Reset gate (r_t)

→ update gate to retain ~~new info~~ old info

→ Reset gate to forget info and rules to take new inputs.

→ Gatting equations for GRU.



(1) $U_t = \sigma\left[W_U\left(h_{t-1} + x_t\right) + b_U\right]$

(2) $r_t = \sigma\left(W_r\left(h_{t-1} + x_t\right) + b_r\right]$

→ ~~update rule~~:

→

Sr. No. 8517

Space For Marks	Question No.	START WRITING HERE (Begin answer for each question on a new page)

→ update rule.

$$\boxed{h_t = u_t \tilde{C} + (1-u_t)h_{t-1}}$$

this can be interpreted as complex combination of the previous hidden state and candidate hidden state.

∴ if the value $u_t = 0$ or 1

if 0.
$h_t = h_{t-1}$
i.e no update needed. use previous state

if 1
$h_t = u_t \tilde{C}$
i.e update the new information.

→ GRU tends to have fewer parameters than LSTM because of number of states & gates
GRU → 1 state & 2 gates
LSTM → 2 state & 3 gates.

~~GRU | LSTM~~

~~① $u_t = \sigma(W_u h_{t-1} + x_t$~~

→

mathematically.

→ GRU

(1) $U_t = \sigma(W_u(\cancel{h}_{t-1} + x_t) + b_u)$
(2) $r_t = \sigma(W_r(h_{t-1} + x_t) + b_r)$
(3) $\tilde{C} = \tanh[W_c(h_{t-1} + x_t) + b_c]$
(4) $h_t = U_t \tilde{C} + (1 - U_t) h_{t-1}$

LSTM

(1) $f_t = \sigma[W_f(h_{t-1} + x_t) + b_f)]$
(2) $i_t = \sigma(W_i(h_{t-1} + x_t) + b_i)$
(3) $O_t = \sigma(W_o(h_{t-1} + x_t) + b_o)$
(4) $\tilde{C} = \tanh[W_c(h_{t-1} + x_t) + b_c$
(5) $C_t = f_t \times C_{t-1} + i_t \times \tilde{C}$
(6) ~~$y_t = O_t \tanh($~~
$y_t = O_t \times \tanh(C_t)$

hence GRU has less parameters

→ This might improve generalisations in real world applications in scenarios like grammar correction applications wherein the model has to store the inputs and accordingly correct ahead whether any grammatical error or not by comparision.

→ & as the sentence to be checked for grammar error could be long term as well as short term.

Sr. No. 8517

Space For Marks	Question No.	START WRITING HERE (Begin answer for each question on a new page)
	Q2b]	~~autoenc~~ under complete autoencoder.

Sr. No. 8517

Space For Marks	Question No.	START WRITING HERE (Begin answer for each question on a new page)

Sr. No. 8517

Space For Marks	Question No.	START WRITING HERE (Begin answer for each question on a new page)

Sr. No. 8517

START WRITING HERE

(Begin answer for each question on a new page)

Space For Marks	Question No.	